

Comprehensive Treatise on IUPAC Electrochemical Cell Conventions, Notations, and Morphological Taxonomy

Rigorous Chemical Nomenclature & Physical Electrochemistry Compendium

September 30, 2026

Abstract

This monograph presents the formal IUPAC (International Union of Pure and Applied Chemistry) standards governing electrochemical cell notation, boundary typologies, and half-cell representations. Every standard, non-standard, and unconventional electrochemical system—spanning metal–ion, insoluble salt, gas, soluble redox, amalgam, insoluble oxide, concentration, membrane/ion-selective, organic redox, and solid-state/molten systems—is cataloged. Crucially, each representation is dissected via a microscopic, bit-by-bit anatomical parsing that explicitly links every vertical line, comma, phase marker, concentration term, and electrode conductor to its underlying IUPAC mandate.

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1 Fundamental IUPAC Rules and Axioms of Cell Notation

The IUPAC Commission on Electrochemistry codified the shorthand representation of galvanic and electrolytic cells to ensure unambiguous reporting of thermodynamic quantities (ΔG , E_{cell}) and interface phenomena.

Axiom 1: Vector of Electron Flow and Spatial Orientation

- **Left-Hand Electrode (Anode / Oxidation site):** The electrode where oxidation occurs spontaneously under standard galvanic discharge is drawn on the **LEFT**.
- **Right-Hand Electrode (Cathode / Reduction site):** The electrode where reduction occurs spontaneously is drawn on the **RIGHT**.
- **External Electron Transit:** Electrons flow spontaneously through the external measurement circuit from **Left to Right** (Anode $\xrightarrow{e^-}$ Cathode).
- **Sign Convention for Cell Potential:** The cell potential E_{cell} is defined by IUPAC as:

$$E_{\text{cell}} = E_{\text{right}} - E_{\text{left}} = E_{\text{cathode}} - E_{\text{anode}} \quad (1)$$

where both E_{right} and E_{left} are standard reduction potentials. A positive E_{cell} denotes thermodynamic spontaneity ($\Delta G = -nFE_{\text{cell}} < 0$) in the direction written from left to right.

Axiom 2: Boundary and Interface Typology

- **Single Vertical Line (|):** Denotes a *phase boundary* across which an interfacial potential difference develops (e.g., solid-liquid, solid-solid, liquid-gas).
- **Double Vertical Line (||):** Denotes a *salt bridge* or electrolyte junction where the liquid junction potential (E_j) has been experimentally eliminated or made thermodynamically negligible ($E_j \approx 0$).
- **Dotted/Dashed Line or Colon (: or |):** Denotes a *liquid-liquid boundary* where two miscible electrolyte solutions are in direct physical contact, with a non-zero, measurable liquid junction potential ($E_j \neq 0$).
- **Comma (,):** Denotes species that exist within the *same continuous phase* (e.g., two soluble redox ions in an aqueous solution, or multiple gaseous species at a common interface).

Axiom 3: Sequence of Species Within Half-Cells

- **Outward-Inward Progression:** The external electrical terminal contacts (metallic current collectors or inert conductors) must occupy the absolute extremities of the diagram (the far left for the anode, the far right for the cathode).
- Moving from the far left inward toward the center corresponds to traversing the anode half-cell from solid conductor $\rightarrow$ intermediate phases $\rightarrow$ electrolyte phase.
- Traversing from the center outward toward the far right corresponds to moving from electrolyte phase $\rightarrow$ intermediate phases $\rightarrow$ solid conductor of the cathode.

Axiom 4: Phase and State Specification

- Every constituent species must be accompanied by its state of aggregation in parentheses: (*s*) for solid, (*l*) for liquid, (*g*) for gas, (*aq*) for aqueous solution, and (amalgam) or (Hg) for metal amalgams.
- For thermodynamic exactness, the activity a_i , molality m_i , molarity c_i , or partial pressure p_i must be explicitly denoted within the parentheses, e.g., $\text{Zn}^{2+}(\text{aq}, a = 0.05)$ or $\text{H}_2(\text{g}, 1.0 \text{ bar})$.

Axiom 5: Role of Inert Electron Collectors

- When a redox half-reaction involves exclusively non-conductive phases (such as dissolved ions, gases, or non-conductive liquids/organics), an inert electronic conductor (Pt, Au, or C(graphite)) must be physically positioned as the terminal phase to provide an electron source/sink.

2 Exhaustive Typology & Bit-by-Bit Analytical Parsing

Below, every conceivable electrochemical cell type is subjected to a microscopic anatomical deconstruction. For each system, we provide:

- The full IUPAC symbolic notation.
- The half-cell and net cell reaction equations.
- A segment-by-segment lexical dissection indicating exactly which IUPAC rule mandates each symbol.

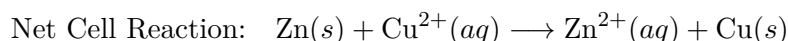
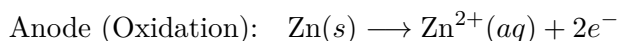
2.1 Category 1: Metal–Metal Ion Electrodes (The Classical Daniell Cell)

System 1.1: Standard Daniell Galvanic Cell

IUPAC Cell Notation:



Half-Cell and Overall Reactions:



Bit-by-Bit IUPAC Rule Parsing:

- **Zn(s):** *Terminal electronic conductor of the anode.* Placed at the extreme left per **Axiom 1** (anode convention) and **Axiom 3** (terminal contact at extremity). State (s) specifies solid state per **Axiom 4**.
- **| (First):** *Solid-Liquid Phase Boundary.* Mandated by **Axiom 2** because the solid zinc electrode is in direct contact with the aqueous electrolyte solution.
- **Zn²⁺(aq, a_{Zn²⁺}):** *Anolyte oxidized species.* Specifies the zinc cations dissolved in aqueous medium with explicit thermodynamic activity $a_{\text{Zn}^{2+}}$ per **Axiom 4**.
- **||:** *Eliminated Junction / Salt Bridge.* Mandated by **Axiom 2**. A potassium chloride or agar-agar bridge prevents physical mixing of bulk electrolytes while keeping liquid junction potential $E_j \approx 0$.
- **Cu²⁺(aq, a_{Cu²⁺}):** *Catholyte reducible species.* Positioned directly to the right of the salt bridge, representing the aqueous ion undergoing reduction.
- **| (Second):** *Liquid-Solid Phase Boundary.* Mandated by **Axiom 2** between the aqueous catholyte and the solid copper electrode.
- **Cu(s):** *Terminal electronic conductor of the cathode.* Positioned at the extreme right per **Axiom 1** (cathode at right) and **Axiom 3** (terminal contact).

2.2 Category 2: Metal–Insoluble Salt–Anion Electrodes (Second-Kind Electrodes)

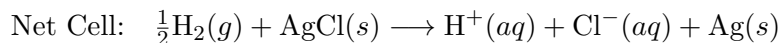
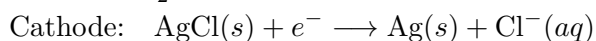
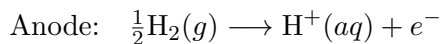
These electrodes feature a metal in direct contact with a sparingly soluble salt of that metal, which is bathed in an electrolyte containing the salt's common anion.

System 2.1: Silver–Silver Chloride Half-Cell Coupled with Standard Hydrogen Electrode

IUPAC Cell Notation:



Half-Cell and Overall Reactions:

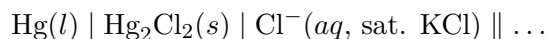


Bit-by-Bit IUPAC Rule Parsing (Cathode Focus):

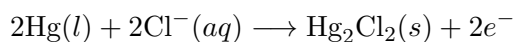
- $\parallel$: *Salt Bridge* separating anolyte and catholyte.
- $\text{Cl}^-(\text{aq}, a_{\text{Cl}^-})$: *Aqueous common anion phase*. Under **Axiom 3**, this solution faces the central cell junction.
- $\mid$: *Solution-Solid Phase Boundary* separating dissolved chloride ions from crystalline solid AgCl.
- $\text{AgCl}(s)$: *Intermediate sparingly soluble salt phase*. Placed between the common anion electrolyte and the base metal.
- $\mid$: *Solid-Solid Interfacial Phase Boundary*. Separates the non-conducting/semiconducting $\text{AgCl}(s)$ crystal coat from metallic silver.
- $\text{Ag}(s)$: *Metallic conductor / Terminal cathode contact*. Placed at the extreme right per **Axiom 1 & 3**.

System 2.2: Saturated Calomel Electrode (SCE) as Anode

IUPAC Half-Cell Notation (as Anode):



Anodic Reaction:



Bit-by-Bit Parsing:

- **Hg(l)**: *Liquid metallic electron conductor*. Placed on the extreme left. Liquid state denoted by (l) per **Axiom 4**.
- **|**: *Liquid-Solid Phase Boundary* between liquid mercury pool and solid mercurous chloride paste.
- **Hg₂Cl₂(s)**: *Insoluble salt layer (calomel)* resting over the mercury.
- **|**: *Solid-Liquid Phase Boundary* between solid calomel paste and the potassium chloride electrolyte.
- **Cl⁻(aq, sat. KCl)**: *Electrolyte solution with fixed chloride activity* saturated with potassium chloride.

2.3 Category 3: Gas Electrodes

Gas electrodes require a catalytic, chemically inert solid phase (typically platinized platinum) to facilitate electron transfer.

System 3.1: Standard Hydrogen Electrode (SHE) coupled with Chlorine Gas Cathode

IUPAC Cell Notation:

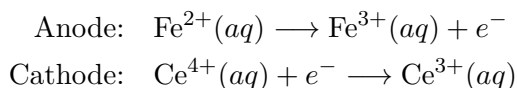


Bit-by-Bit IUPAC Rule Parsing:

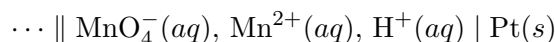
- **Pt(s)** (Far Left): *Inert conductor*. Mandated by **Axiom 5** because $\text{H}_2(\text{g})$ cannot serve as a rigid electron terminal.
- **|**: *Solid-Gas Boundary* where platinum is bathed in hydrogen gas.
- **H₂(g, p_{H₂)}**: *Gaseous reactant* with defined partial pressure p in bar.
- **|**: *Gas-Liquid Boundary* where H_2 gas interfaces with aqueous hydronium ions.
- **H⁺(aq, a_{H⁺)}**: *Hydronium ion anolyte*.
- **||**: *Salt Bridge*.
- **Cl⁻(aq, a_{Cl⁻)}**: *Chloride ion catholyte*.
- **|**: *Liquid-Gas Boundary*.
- **Cl₂(g, p_{Cl₂)}**: *Gaseous oxidant*.
- **|**: *Gas-Solid Boundary*.
- **Pt(s)** (Far Right): *Inert cathode current collector*.

2.4 Category 4: Homogeneous Redox Electrodes

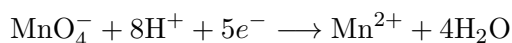
Systems where both reduced and oxidized forms reside within the same homogeneous phase.

System 4.1: Iron(II)/Iron(III) coupled with Cerium(IV)/Cerium(III)**IUPAC Cell Notation:****Reactions:****Bit-by-Bit IUPAC Rule Parsing:**

- **Pt(s)**: *Inert electron sink*. Essential per **Axiom 5**.
- **|**: *Phase boundary* between solid platinum and the single aqueous solution.
- **Fe²⁺(aq), Fe³⁺(aq)**: *Homogeneous redox couple separated by a COMMA*. Mandated by **Axiom 2**; using a vertical line **|** between Fe²⁺ and Fe³⁺ is strictly incorrect because both ions are co-dissolved in the exact same water phase.
- **||**: *Salt Bridge*.
- **Ce⁴⁺(aq), Ce³⁺(aq)**: *Catholyte redox couple separated by a COMMA* for identical phase continuity reasons.
- **| Pt(s)**: *Phase boundary* to inert platinum cathode collector.

System 4.2: Acidic Permanganate Multi-Species Redox System**IUPAC Half-Cell Notation (Cathodic Reduction):****Bit-by-Bit Parsing:**

- **MnO₄⁻(aq), Mn²⁺(aq), H⁺(aq)**: All three active chemical participants are co-present in the same aqueous solution. Under **Axiom 2**, all are separated by **commas**. Protons H⁺ must be listed because they participate directly in the stoichiometry:

**2.5 Category 5: Metal Amalgam Electrodes**

Amalgams consist of reactive metals dissolved in liquid mercury. They allow electrochemical interrogation of alkali/alkaline-earth metals without spontaneous explosive reactions with water.

System 5.1: Sodium Amalgam Concentration Cell

IUPAC Cell Notation:



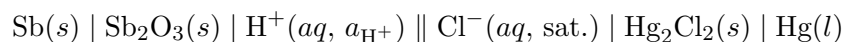
Bit-by-Bit IUPAC Rule Parsing:

- **Na(amalgam, a_1)**: Solid/liquid alloy of sodium dissolved in mercury at activity a_1 . Acts as its own conductor. Placed at the anode extremity.
- **|**: *Phase boundary* between metal amalgam phase and aqueous electrolyte.
- **Na⁺(aq, c)**: Common aqueous electrolyte uniting both half-cells. Notice: *no salt bridge* (||) is present because both amalgam electrodes dip into the **same** electrolyte solution.
- **|**: *Phase boundary* from common electrolyte into cathodic amalgam.
- **Na(amalgam, a_2)**: Cathodic sodium amalgam at lower thermodynamic activity a_2 . Spontaneous transfer proceeds from high activity to low activity.

2.6 Category 6: Insoluble Metal Oxide / Hydroxide Electrodes

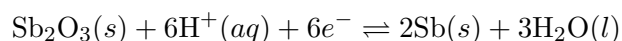
System 6.1: Antimony pH-Sensing Electrode coupled with Saturated Calomel

IUPAC Cell Notation:

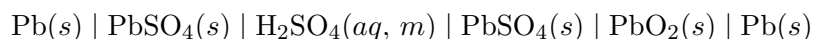


Bit-by-Bit Parsing (Anode):

- **Sb(s)**: Pure metalloid antimony serving as structural base and electronic conductor.
- **|**: *Phase boundary* between elemental antimony and its solid oxide crust.
- **Sb₂O₃(s)**: Insoluble oxide layer participating in the hydration/dissolution equilibrium:



- **|**: *Solid-Liquid boundary* into the analyte solution whose pH is being determined.

System 6.2: Complete Lead-Acid Secondary Cell (Discharge State)**IUPAC Cell Notation:****Bit-by-Bit Parsing:**

- **Pb(s)** (Left): Lead sponge anode current collector.
- **| PbSO₄(s)**: Insoluble lead sulfate formed on anode during oxidation.
- **| H₂SO₄(aq, m)**: Shared aqueous sulfuric acid electrolyte. Single lines on both sides because both electrodes share the same bulk liquid without a diaphragm.
- **PbSO₄(s) |**: Lead sulfate precipitation layer on cathode.
- **PbO₂(s)**: Insoluble lead(IV) oxide depolarizer.
- **| Pb(s)** (Right): Supporting lead grid current collector serving as terminal cathode contact.

2.7 Category 7: Concentration Cells

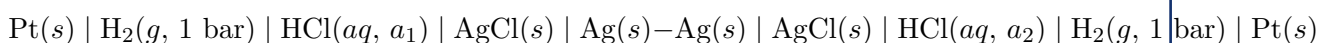
Concentration cells derive their electromotive force solely from differences in activity or partial pressure.

System 7.1: Electrolyte Concentration Cell WITH Transference (Liquid Junction)**IUPAC Cell Notation:****Bit-by-Bit Parsing:**

- **Ag(s)** (Left): Anodic silver electrode undergoing dissolution ($\text{Ag} \rightarrow \text{Ag}^+ + e^-$).
- **| AgNO₃(aq, a₁)**: Anolyte of lower activity.
- **:** (Colon / Dashed Line): *Direct Liquid-Liquid Junction with Transference*. Mandated by **Axiom 2**. Ions freely migrate across this boundary governed by their transport numbers (t_+, t_-), establishing a finite liquid junction potential E_j .
- **AgNO₃(aq, a₂)**: Catholyte of higher activity.
- **| Ag(s)** (Right): Cathodic silver electrode undergoing deposition ($\text{Ag}^+ + e^- \rightarrow \text{Ag}$).

System 7.2: Electrolyte Concentration Cell WITHOUT Transference

Two independent cells connected back-to-back in electrical series opposition:

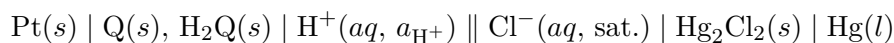


Here, the physical hyphen (–) represents the direct metallic coupling between the two silver phases, completely preventing ionic mixing. Hence, $E_j = 0$ rigorously.

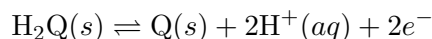
2.8 Category 8: Organic and Biological Redox Couples

System 8.1: The Quinhydrone Half-Cell (Coupled with Calomel)

IUPAC Cell Notation:



Anodic Reaction:



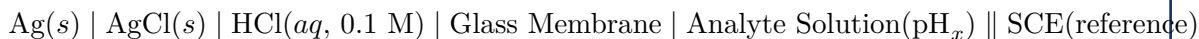
Bit-by-Bit Parsing:

- **Pt(s)**: Inert terminal collector for electron exchange.
- **|**: Phase boundary to organic crystals.
- **Q(s), H₂Q(s)**: Benzoquinone (Q) and hydroquinone (H₂Q) present as equimolar solid adducts. Separated by a **comma** because they co-exist intimately in the same solid crystalline phase (the 1 : 1 quinhydrone molecular complex).
- **|**: Phase boundary to aqueous buffer containing the hydronium ion species.

2.9 Category 9: Membrane and Ion-Selective Electrodes (ISE)

System 9.1: Complete Glass-Electrode pH Measurement Cell

IUPAC Cell Notation:



Bit-by-Bit Parsing:

- **Ag(s) | AgCl(s) | HCl(aq, 0.1 M)**: Internal primary reference electrode with fixed chloride activity.
- **| Glass Membrane |**: The hydrated aluminosilicate or lithia glass matrix. The two vertical lines flanking the membrane represent the two distinct phase boundaries on the inner and outer surfaces where cation exchange ($\text{Na}^+ \rightleftharpoons \text{H}^+$) occurs.
- **Analyte Solution(pH_x)**: The unmeasured sample solution possessing unknown hydronium ion activity.
- **|| SCE**: Porous ceramic junction and saturated calomel external reference completing the measurement circuit.

2.10 Category 10: Advanced Solid-State & Molten Industrial Systems

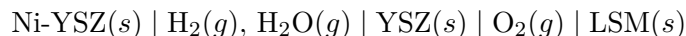
System 10.1: Modern Lithium-Ion Battery (Intercalation Cell in Charged State)

IUPAC Cell Notation:



Bit-by-Bit Parsing:

- **Cu(s)**: Copper negative current collector terminal.
- **|**: Phase boundary between Cu foil and graphitic anode coating.
- **Li_xC₆(s)**: Lithiated graphite intercalation compound.
- **|**: Solid-Liquid phase boundary into non-aqueous electrolyte.
- **LiPF₆(EC:DMC)**: Lithium hexafluorophosphate dissolved in ethylene carbonate / dimethyl carbonate organic blend.
- **|**: Liquid-Solid phase boundary into cathode coating.
- **Li_{1-x}CoO₂(s)**: Delithiated lithium cobalt oxide intercalation material.
- **|**: Cathode coating to aluminum foil boundary.
- **Al(s)**: Aluminum positive current collector terminal.

System 10.2: Solid Oxide Fuel Cell (SOFC)**IUPAC Cell Notation:****Bit-by-Bit Parsing:**

- **Ni – YSZ(s)**: Porous nickel–yttria-stabilized zirconia cermet serving as both anode catalyst and electronic conductor.
- **|**: Solid-Gas boundary where fuel gas streams over the porous anode.
- **H₂(g), H₂O(g)**: Gaseous fuel and combustion product coexisting in the anode flow channel (separated by a **comma**).
- **|**: Anode gas channel to solid electrolyte phase boundary.
- **YSZ(s)**: Dense Yttria-Stabilized Zirconia ceramic operating as a purely O²⁻ oxygen-ion conducting solid electrolyte.
- **|**: Solid electrolyte to cathode gas chamber boundary.
- **O₂(g)**: Oxygen/air oxidant stream.
- **|**: Gas to porous solid cathode boundary.
- **LSM(s)**: Strontium-doped Lanthanum Manganite (La_{1-x}Sr_xMnO₃) ceramic cathode current collector.

3 Master Reference Summary Matrix

Electrode Category	Structural Scheme	Canonical Example Notation
Metal/Metal Ion	$M(s) \mid M^{n+}(aq)$	$Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$
Insoluble Salt	$M(s) \mid MX(s) \mid X^{-}(aq)$	$Ag(s) \mid AgCl(s) \mid Cl^{-}(aq) \parallel \dots$
Gas Electrode	$Pt(s) \mid Gas(g) \mid Ion(aq)$	$Pt(s) \mid H_2(g, p) \mid H^{+}(aq, a)$
Soluble Redox	$Pt(s) \mid Red(aq), Ox(aq)$	$Pt(s) \mid Fe^{2+}(aq), Fe^{3+}(aq)$
Amalgam	$M(Hg) \mid M^{n+}(aq)$	$Na(Hg, a_1) \mid Na^{+}(aq) \mid Na(Hg, a_2)$
Insoluble Oxide	$M(s) \mid MO_x(s) \mid H^{+}(aq)$	$Sb(s) \mid Sb_2O_3(s) \mid H^{+}(aq)$
Liquid Junction	$Soln_1 : Soln_2$	$Ag(s) \mid AgNO_3(a_1) : AgNO_3(a_2) \mid Ag(s)$
ISE Membrane	$Soln_{int} \mid Membrane \mid Soln_{ext}$	$HCl(0.1\ M) \mid Glass \mid Analyte(pH_x)$
Solid-State Inter-cal.	$Collector \mid Host(s) \mid Electrolyte$	$Cu(s) \mid Li_xC_6(s) \mid LiPF_6 \mid Li_{1-x}CoO_2(s) \mid Al(s)$

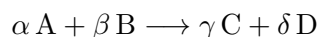
Table 1: Comprehensive IUPAC Electrochemical System Classification Matrix.

4 Bidirectional Translation Heuristics: Reaction $\longleftrightarrow$ Notation

Students frequently stumble when moving between chemical equations and IUPAC cell notation because they attempt to memorize individual cells rather than applying a systematic, mechanical algorithm. Even for unfamiliar, non-stoichiometric, or intricate systems (e.g., lithium intercalation or multi-electron fuel cells), the exact IUPAC cell notation can be deduced using two deterministic conversion algorithms.

4.1 Algorithm 1: Net Cell Reaction $\longrightarrow$ IUPAC Cell Notation

Given an overall balanced reaction:



1. **Assign Oxidation Numbers and Bisect the Reaction:** Identify which elements increase in oxidation state (oxidation / anode) and which decrease (reduction / cathode). Separate into two independent half-cell equations:



2. **Identify Physical Phases and Terminal Conductors:** Check each half-reaction for the presence of an electrically conductive solid metal (M(s)):

- If a metallic solid is present, it serves as the terminal current collector at the extremity.
- If all species are fluid (aqueous ions, gases, liquid organics) or non-conductive solids/semiconductors, introduce an inert current collector (typically Pt(s) or C(graphite)) at that half-cell's outer edge.

3. **Construct Spatial Sequences (Anode at Left, Cathode at Right):**

- **Left Half-Cell (Oxidation):** Write the terminal conductor on the far left. Move rightward toward the bridge/interface:

Terminal Conductor | Intermediate Solid/Gas Phases | Electrolyte Phase

- **Right Half-Cell (Reduction):** Move from the central bridge outward to the far right:

Electrolyte Phase | Intermediate Solid/Gas Phases | Terminal Conductor

4. **Apply IUPAC Delimiter Rules Strictly:**

- Use | only across distinct phases (e.g., solid/solution, solid/gas).
- Use a comma (,) between species sharing the identical continuous phase (e.g., ions in common solution).
- Use || for a salt bridge, or a single shared phase if no membrane separates them (e.g., secondary batteries sharing a common acid/base bath).

4.2 Algorithm 2: IUPAC Cell Notation $\longrightarrow$ Half-Reactions & Net Cell Reaction

Given a cell notation string:

Phase 1 | Phase 2 || Phase 3 | Phase 4

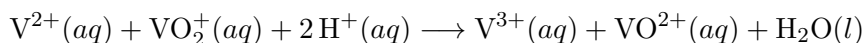
1. **Split at the Central Junction:** Everything to the left of $\parallel$ (or the central electrolyte) constitutes the **Anode** (e^- produced). Everything to the right constitutes the **Cathode** (e^- consumed).
2. **Ignore Inert Conductors in Mass Balance:** If $\text{Pt}(s)$, $\text{Au}(s)$, or $\text{C}(s)$ is present solely as an electron sink, cross it out from stoichiometric mass balancing; it transfers electrons without appearing in the chemical formula equation.
3. **Formulate Anodic Half-Reaction:** Write the species on the far left transforming into the species toward the center. Balance mass, oxygen/hydrogen (using H_2O , H^+ , or OH^- depending on medium), and add $n e^-$ to the product (right) side.
4. **Formulate Cathodic Half-Reaction:** Write the species near the center transforming into the species toward the far right. Balance mass, oxygen/hydrogen, and add $n e^-$ to the reactant (left) side.
5. **Equalize Electrons and Sum:** Multiply half-reactions by suitable integers such that $n e^-$ cancels out completely, yielding the net cell reaction.

5 Comprehensive Graded Practice Problem Set

5.1 Part I: Deducing IUPAC Cell Notation from Net Reactions

Problem 1.1: Vanadium Redox Flow Battery (All-Liquid Soluble Ions)

Given Reaction: The positive and negative electrolytes of a vanadium flow battery undergo the following net discharge reaction in an acidic aqueous medium (H_2SO_4):



Deduce the complete IUPAC cell notation, assuming an ion-exchange membrane acts as the central separator.

Step-by-Step Derivation:

1. Oxidation States & Half-Reactions:

Anode: $\text{V}^{2+}(\text{aq}) \longrightarrow \text{V}^{3+}(\text{aq}) + e^-$ (V increases from +2 $\rightarrow$ +3)

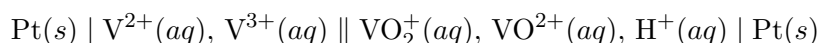
Cathode: $\text{VO}_2^+(\text{aq}) + 2\text{H}^+(\text{aq}) + e^- \longrightarrow \text{VO}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ (V decreases from +5 $\rightarrow$ +4)

2. *Conductor Requirements:* All active vanadium and proton species are dissolved in water. Neither side contains solid metal. Therefore, both electrodes require an inert conductor: C(graphite) or Pt(s).

3. Phase Boundaries & Commas:

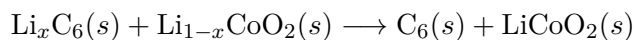
- Anode solution contains V^{2+} and V^{3+} . Since they are in the same phase, separate by a **comma**: $\text{V}^{2+}(\text{aq}), \text{V}^{3+}(\text{aq})$.
- Cathode solution contains VO_2^+ , H^+ , and VO^{2+} . Co-dissolved in one phase, separated by **commas**: $\text{VO}_2^+(\text{aq}), \text{VO}^{2+}(\text{aq}), \text{H}^+(\text{aq})$.

4. Final IUPAC Notation:

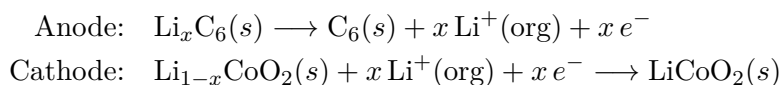


Problem 1.2: Complex Non-Stoichiometric Lithium-Ion Intercalation Cell

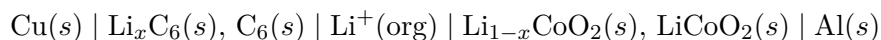
Given Reaction: During discharge of a commercial lithium-ion battery, lithium ions de-intercalate from a layered graphite host and intercalate into a cobalt oxide lattice:



The cell operates using a non-aqueous electrolyte containing dissolved Li^+ ions (e.g., LiPF_6 in organic carbonate). The negative terminal uses copper foil (Cu), and the positive terminal uses aluminum foil (Al). Deduce the IUPAC notation.

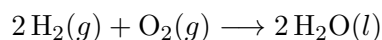
Step-by-Step Derivation:1. *Oxidation States & Half-Reactions:*2. *Spatial Arrangement:*

- Far Left (Anode Terminal): $\text{Cu}(s)$ current collector.
- Anode Active Layer: Solid mixture of lithiated graphite $\text{Li}_x\text{C}_6(s)$ and delithiated $\text{C}_6(s)$ (separated by a comma or written as the active intercalation solid phase).
- Electrolyte: Non-aqueous liquid containing $\text{Li}^+(\text{org})$.
- Cathode Active Layer: Intercalation oxide $\text{Li}_{1-x}\text{CoO}_2(s)$, $\text{LiCoO}_2(s)$.
- Far Right (Cathode Terminal): $\text{Al}(s)$ current collector.

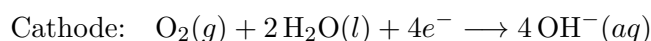
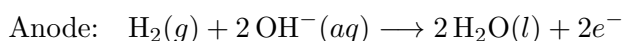
3. *Final IUPAC Notation:*

Problem 1.3: Alkaline Fuel Cell (AFC) Consuming Gas Phases

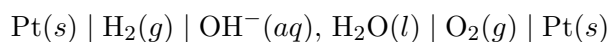
Given Reaction: Gaseous hydrogen and oxygen react in a concentrated aqueous potassium hydroxide electrolyte:



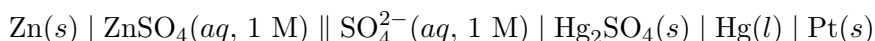
Deduce the IUPAC representation.

Step-by-Step Derivation:1. *Alkaline Half-Reactions:*

2. *Electrode Boundaries:* Both gases require porous catalytic inert substrates (e.g., Pt(s) or Ni(s)). Both half-cells contact the same bulk aqueous alkaline solution (OH[−]).

3. *Final IUPAC Notation:*

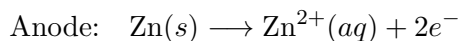
(Alternatively, with a porous matrix separating chambers: Pt(s) | H₂(g) | OH[−](aq) || OH[−](aq) | O₂(g) | Pt(s))

5.2 Part II: Deducing Half-Reactions and Net Reactions from Cell Notation**Problem 2.1: Mercury–Mercurous Sulfate Reference Cell****Given Cell Notation:**

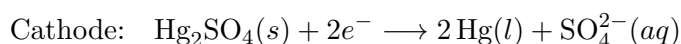
Deduce: (i) The anodic half-reaction, (ii) The cathodic half-reaction, and (iii) The net cell reaction.

Step-by-Step Derivation:

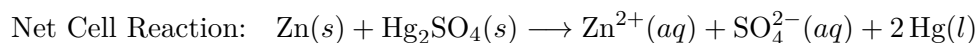
1. *Anode (Left of ||):* Species present: Zn(s) and Zn²⁺(aq). Oxidation of zinc metal:

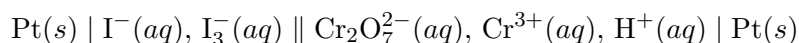


2. *Cathode (Right of ||):* Species present: SO₄^{2−}(aq), solid Hg₂SO₄(s), and liquid mercury Hg(l). Note that Pt(s) is merely a contact lead dipping into the mercury pool. Reduction converts the mercurous salt into elemental mercury:



3. *Sum of Half-Reactions (2e[−] cancel directly):*

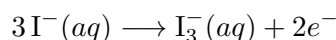


Problem 2.2: Dichromate–Halide Mixed Phase Redox System**Given Cell Notation:**

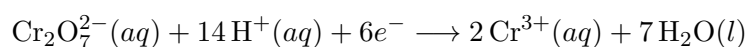
Deduce the balanced half-reactions and the stoichiometric overall cell reaction.

Step-by-Step Derivation:

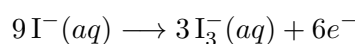
1. *Anode Half-Reaction (Left):* Iodide ions (I^-) are oxidized to triiodide ions (I_3^-):



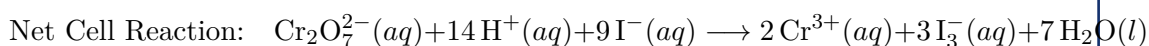
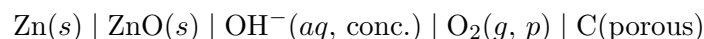
2. *Cathode Half-Reaction (Right):* Dichromate ($\text{Cr}_2\text{O}_7^{2-}$) is reduced to chromium(III) (Cr^{3+}) in an acidic solution:



3. *Equalizing Electrons & Net Reaction:* Multiply the anodic equation by 3 to balance $6e^-$:



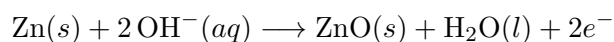
Adding the equations yields:

**Problem 2.3: Alkaline Zinc–Air Battery System****Given Cell Notation:**

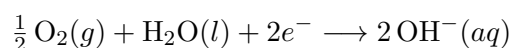
Deduce the half-reactions and determine whether the overall cell consumption involves water.

Step-by-Step Derivation:

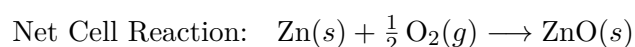
1. *Anode Half-Reaction (Left):* Zinc metal oxidizes in presence of hydroxide to form solid zinc oxide precipitate:



2. *Cathode Half-Reaction (Right):* Atmospheric oxygen is reduced at the porous carbon gas-diffusion electrode:



3. *Summation* ($2e^-$, 2OH^- , and H_2O cancel completely):



Conclusion: Water acts as an internal intermediate but does not appear in the net reaction!